A.3 JSX best practices

JSX is a fundamental aspect of React that blends the flexibility of JavaScript with an HTML-like syntax for defining user interfaces. Well-structured JSX enhances code readability, streamlines maintenance, and minimizes bugs—qualities highly valued in technical interviews when evaluating React proficiency.

This appendix presents essential JSX best practices to help developers write clean, maintainable, and efficient React code. The following sections provide practical recommendations for structuring components, applying proper syntax and styling, and managing whitespace effectively.

A.3.1 Understanding JSX Transpilation

The JSX code `const element = <h1>Hello, World!</h1>` is transpiled into the JavaScript: `const element = React.jsx('h1', { children: 'Hello, World!' })`.

This transformation allows browsers to execute JSX as valid JavaScript. To see this process in action, you can disable JSX development mode in the Vite application by updating `vite.config.ts` as shown in listing A.1.

Listing A.1 Configuring Vite to disable JSX development mode

```
import { defineConfig } from 'vite';
import react from '@vitejs/plugin-react';
// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
  esbuild: {
    jsxDev: false, #A
  },
});
```

After disabling JSX development mode, inspect the transpiled code in the browser's Network tab. For example, the original `main.tsx` file (listing A.2) is transformed into its transpiled version (listing A.3).